

The image shows a Visual Studio Code editor window with a project named `AI_LAB_PROGRAMS`. The Explorer sidebar on the left lists the project files, including `representationofknowledgeusingpredicatelogicandrulebasedsystems.py`, which is the active file. The editor displays the following Python code:

```
45         'Socrates_is_human',
46         'All_humans_are_mortal'
47     ],
48     'then': 'Socrates_is_mortal'
49 }
50
51
52 # create Rule-Based System
53 rbs = RuleBasedSystem(initial_facts, production_rules)
54
55 print("Initial Facts:", initial_facts)
56
57 # Perform forward chaining
58 final_kb = rbs.forward_chain()
59
60 print("Final Knowledge Base Facts:", final_kb)
```

The TERMINAL panel at the bottom shows the output of running the script:

```
PS C:\Users\keziy\OneDrive\Desktop\AI_LAB_PROGRAMS> python -u "c:\Users\keziy\OneDrive\Desktop\AI_LAB_PROGRAMS\representationofknowledgeusingpredicatelogicandrulebasedsystems.py"
Initial Facts: ['Socrates_is_human', 'All_humans_are_mortal']
Rule Triggered: IF ['Socrates_is_human', 'All_humans_are_mortal'] THEN Add Socrates_is_mortal
Final Knowledge Base Facts: {'Socrates_is_human', 'Socrates_is_mortal', 'All_humans_are_mortal'}
PS C:\Users\keziy\OneDrive\Desktop\AI_LAB_PROGRAMS>
```

The status bar at the bottom indicates the current file is `main*` at line 60, column 51, using Python 3.10 (64-bit).